

AYUSH GUPTA

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with extensive experience designing and scaling core product architectures, high-performance user interfaces, and robust backend systems. Proven track record of **leading engineering initiatives**, optimizing frontend build pipelines, and driving code quality standards in fast-paced environments. Expert in **React, TypeScript, and Go**, with deep hands-on experience building complex graph-based workflow engines, data-dense virtualized applications, and cloud-deployed real-time systems.

SKILLS

Languages & Frameworks: TypeScript, JavaScript (ES6+), Go (Golang), Java (8/11/17), React, Next.js, Spring Boot
State & Architecture: WebSockets, REST APIs, Zustand, Redux, Context API, Hibernate/JPA
Databases: PostgreSQL, MySQL
Tools & Build Systems: Vite, Create React App (CRA), Docker, Maven, Git
Cloud & DevOps: AWS (EC2, RDS, S3, Lambda, CloudWatch, SNS), Cloudflare, CI/CD (Git-based)

RELEVANT EXPERIENCE

Busibud Solutions Pvt.Ltd. September 2023 - Present

Software Developer (Frontend Lead)

- Architected a **graph-based workflow automation engine (DAG)** utilizing custom topological sorting algorithms for dynamic dependency resolution and cycle-detection within complex execution chains.
- Engineered a high-performance **spreadsheet engine using Zustand and DOM virtualization**, leveraging atomic state selectors to isolate cell-level re-renders during real-time event streams and restricting rendering to the active viewport to minimize memory overhead.
- Designed a cascading **caching strategy for formula evaluations and cross-sheet dependencies**, optimizing state propagation and preventing redundant tree traversals when cell mutations triggered reference updates or filters.
- Led the frontend engineering division of 5 developers for 1.5 years**, taking ownership of release cycles, mentoring engineers, and driving migrations from legacy CRA to Vite to reduce local build times by 50% across multiple standalone repositories.
- Spearheaded a **backend migration project to Java (Spring Boot)**, designing scalable RESTful APIs, Hibernate/JPA data access layers, and containerized Docker services integrated with Git-based CI/CD pipelines for AWS deployment.

EDUCATION

Institute of Engineering and Technology, Agra (DBRAU) 2019 - 2023
Bachelor of Engineering, Computer Science

PROJECTS

Cloud-Deployed Real-Time Chat System - [Link](#)

Go, React, TypeScript, PostgreSQL, Docker, AWS (EC2/RDS), Cloudflare, WebSockets

- Designed and built a highly concurrent, real-time messaging server in Go** utilizing WebSockets for duplex communication and standard REST endpoints for user authentication and room management.
- Implemented persistent data storage by modeling** and optimizing transactional schemas in PostgreSQL, deployed via AWS RDS to ensure reliable message delivery.
- Configured end-to-end cloud infrastructure**, deploying the containerized Go application on an AWS EC2 instance **managed behind a Cloudflare proxy with custom subdomain routing**.